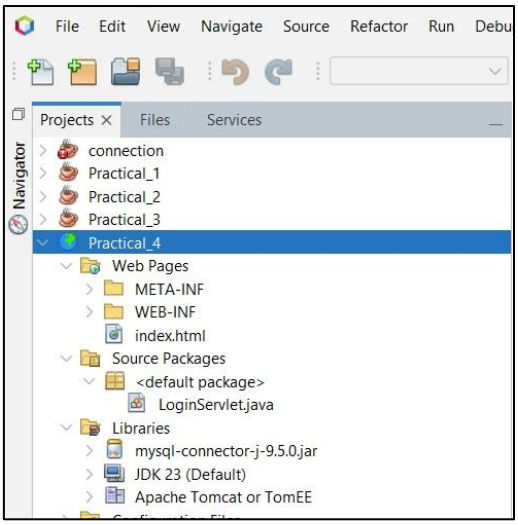
PRACTICAL 4

Develop a data driven servlet application for user authentication. Also display header values of request object and response object.



CODE:

index.html

```
<!DOCTYPE html>
<html>
<head>
   <title>User Login</title>
</head>
<body>
   <h2>User Authentication - Practical 4</h2>

   <form action="LoginServlet" method="post">
      Username: <input type="text" name="username"><br><br>
      Password: <input type="password" name="password"><br><br>
      <input type="submit" value="Login">
```

```
    </form>
</body>
</html>
```

LoginServlet.java

```
import java.io.*;
import java.sql.*;
import java.util.Enumeration;
import javax.servlet.*;
import javax.servlet.http.*;
import javax.servlet.annotation.WebServlet;

@WebServlet("/LoginServlet")
public class LoginServlet extends HttpServlet {

    protected void doPost(HttpServletRequest request, HttpServletResponse
response)
            throws ServletException, IOException {

        response.setContentType("text/html");
        PrintWriter out = response.getWriter();

        String username = request.getParameter("username");
        String password = request.getParameter("password");

        try {
            // Load JDBC Driver
```

```
Class.forName("com.mysql.cj.jdbc.Driver");

// Create Connection
Connection con = DriverManager.getConnection(
    "jdbc:mysql://localhost:3306/practical_4",
    "root", ""
);
// SQL Query
PreparedStatement ps = con.prepareStatement(
    "SELECT * FROM user WHERE username=? AND password=?"
);
ps.setString(1, username);
ps.setString(2, password);

ResultSet rs = ps.executeQuery();

// Authentication Logic
if (rs.next()) {
    out.println("<h2>Login Successful</h2>");
} else {
    out.println("<h2>Invalid Username or Password</h2>");
}
// Request Headers
out.println("<h3>Request Header Values</h3>");
Enumeration<String> reqHeaders = request.getHeaderNames();
while (reqHeaders.hasMoreElements()) {
```

```
            String h = reqHeaders.nextElement();
            out.println(h + " : " + request.getHeader(h) + "<br>");
        }
        // Response Headers
        out.println("<h3>Response Header Values</h3>");
        response.setHeader("Application", "Practical_4");

        for (String h : response.getHeaderNames()) {
            out.println(h + " : " + response.getHeader(h) + "<br>");
        }

        // Close Resources
        rs.close();
        ps.close();
        con.close();

    } catch (ClassNotFoundException e) {
        out.println("Error: JDBC Driver not found<br>");
        out.println(e);
    } catch (SQLException e) {
        out.println("Error: Database problem<br>");
        out.println(e);
    }}}
```

For Database: *practical_4*

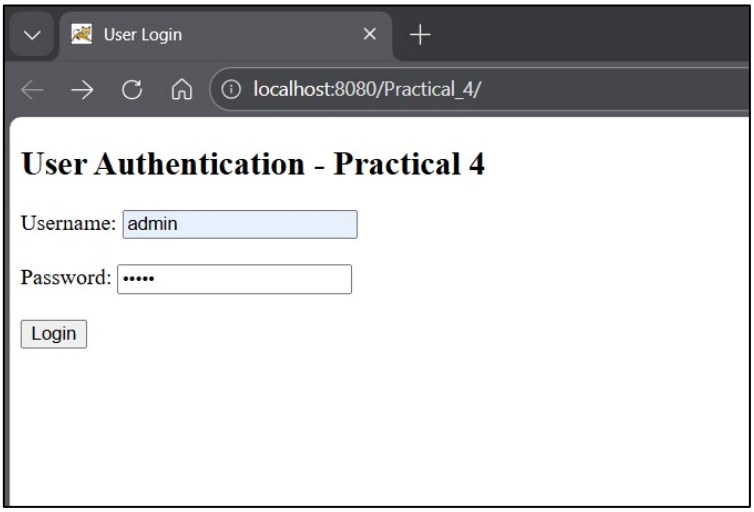
For table:

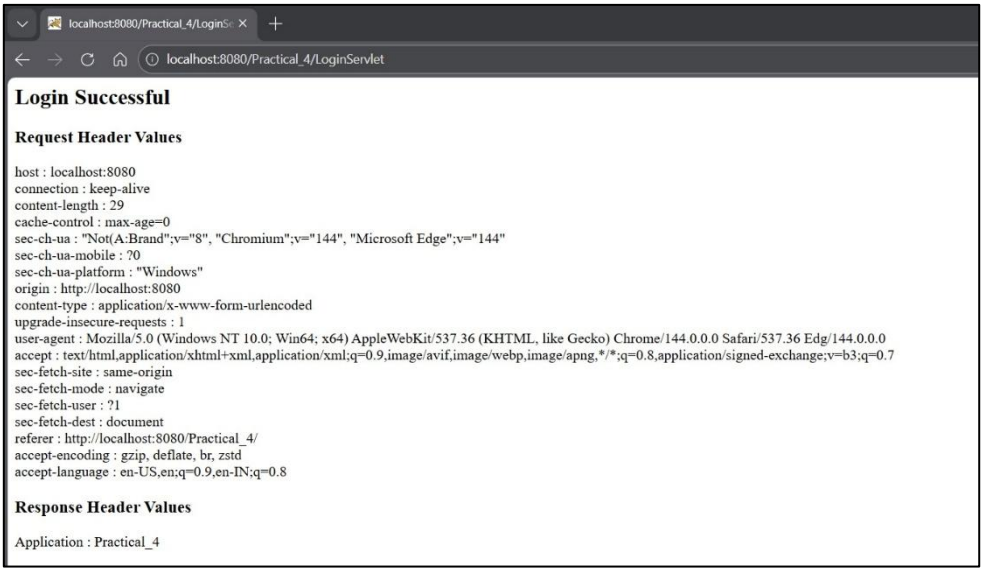
Create user Table

(with Auto Increment Primary Key)

```
CREATE TABLE user (
   id INT AUTO_INCREMENT PRIMARY KEY,
   username VARCHAR(50) NOT NULL,
   password VARCHAR(50) NOT NULL
);
INSERT INTO user (username, password)
VALUES ('admin', '123');
```

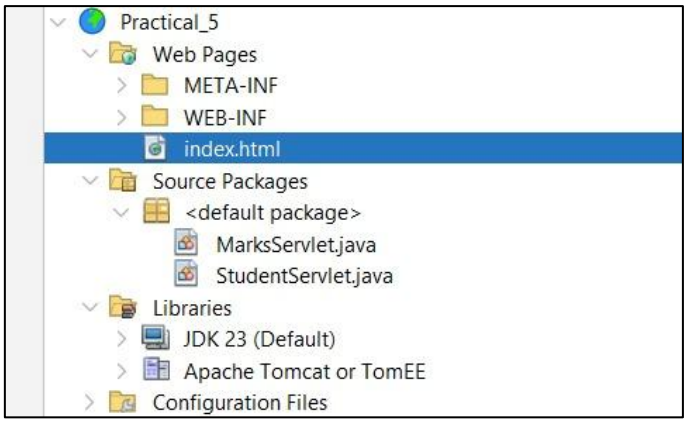
OUTPUT:





PRACTICAL 5

Develop a servlet-based student grading system where the student enters personal details on the first page, marks for six subjects on the second page, and views all details with a color-coded Pass/Fail result on the final page. The student's name should be displayed on every page using session management.



CODE:

index.html

```
<!DOCTYPE html>
<html>
<head>
   <title>Student Details</title>
</head>
<body>
   <h2>Student Information</h2>

   <form action="StudentServlet" method="post">
      Name: <input type="text" name="name"><br><br>
      ID: <input type="text" name="id"><br><br>
```

```
        Department: <input type="text" name="dept"><br><br>


        <input type="submit" value="Next">
    </form>
</body>
</html>
```

MarksServlet.java

```
import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;
import javax.servlet.annotation.WebServlet;

@WebServlet("/MarksServlet")
public class MarksServlet extends HttpServlet {

    protected void doPost(HttpServletRequest request, HttpServletResponse
response)
            throws ServletException, IOException {

        response.setContentType("text/html");
        PrintWriter out = response.getWriter();

        HttpSession session = request.getSession();

        String name = (String) session.getAttribute("name");
        String id = (String) session.getAttribute("id");
```

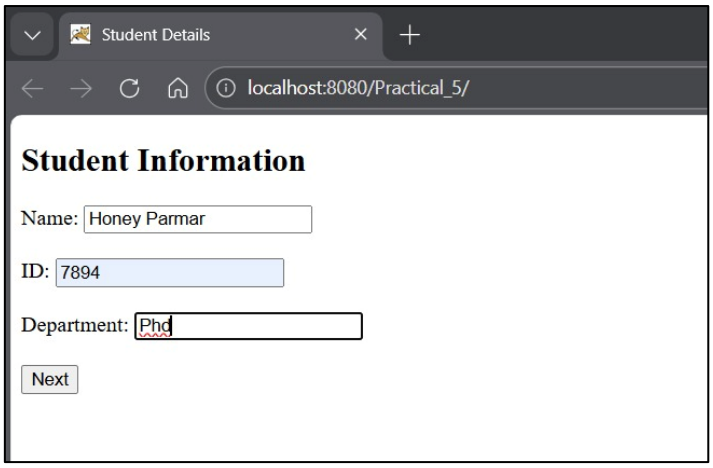
```
        String dept = (String) session.getAttribute("dept");

        int m1 = Integer.parseInt(request.getParameter("m1"));
        int m2 = Integer.parseInt(request.getParameter("m2"));
        int m3 = Integer.parseInt(request.getParameter("m3"));
        int m4 = Integer.parseInt(request.getParameter("m4"));
        int m5 = Integer.parseInt(request.getParameter("m5"));
        int m6 = Integer.parseInt(request.getParameter("m6"));

        boolean pass = m1>=35 && m2>=35 && m3>=35 &&
                    m4>=35 && m5>=35 && m6>=35;

        out.println("<h2>Result</h2>");
        out.println("<h3>Student Name: " + name + "</h3>");
        out.println("ID: " + id + "<br>");
        out.println("Department: " + dept + "<br><br>");

        if (pass) {
            out.println("<h2 style='color:green'>PASS</h2>");
        } else {
            out.println("<h2 style='color:red'>FAIL</h2>");
        }

        session.invalidate();
    }
}
```

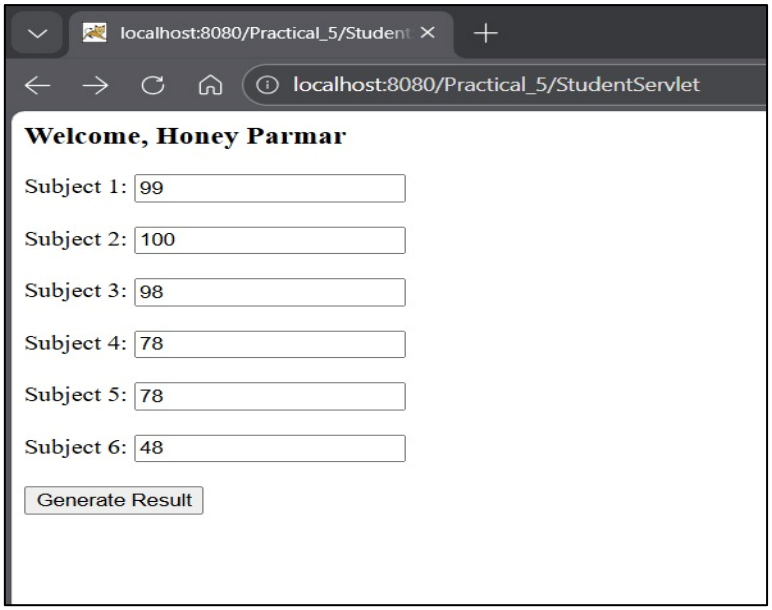
StudentServlet.java

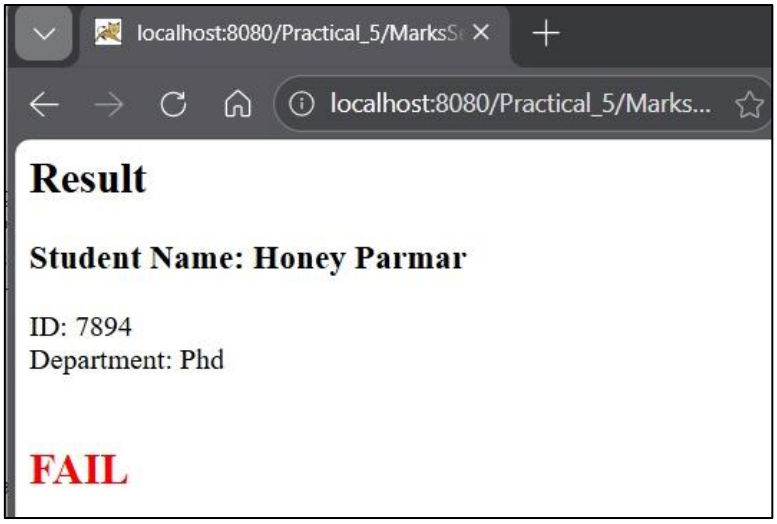
```
import java.io.*;

import javax.servlet.*;

import javax.servlet.http.*;

import javax.servlet.annotation.WebServlet;


@WebServlet("/StudentServlet")

public class StudentServlet extends HttpServlet {


   protected void doPost(HttpServletRequest request, HttpServletResponse
response)

         throws ServletException, IOException {


      response.setContentType("text/html");

      PrintWriter out = response.getWriter();


      String name = request.getParameter("name");

      String id = request.getParameter("id");

      String dept = request.getParameter("dept");


      HttpSession session = request.getSession();

      session.setAttribute("name", name);

      session.setAttribute("id", id);

      session.setAttribute("dept", dept);


      out.println("<h3>Welcome, " + name + "</h3>");

      out.println("<form action='MarksServlet' method='post'>");
```

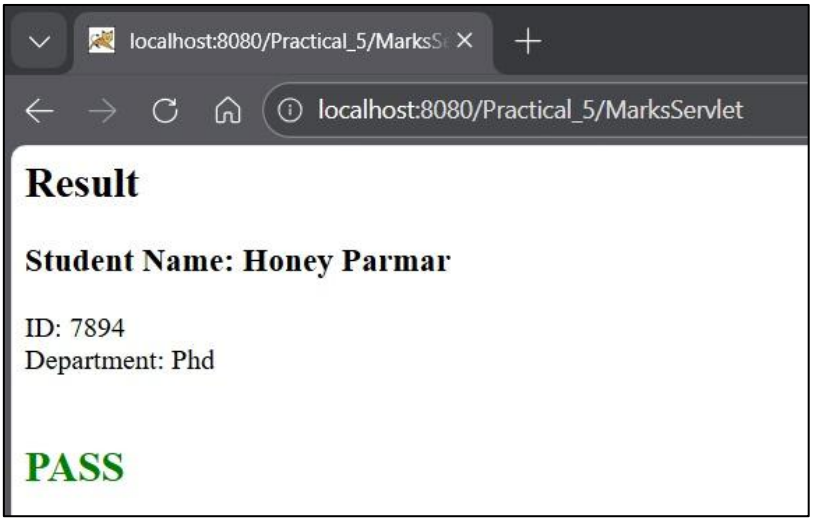
```
        out.println("Subject 1: <input type='number' name='m1'><br><br>");
        out.println("Subject 2: <input type='number' name='m2'><br><br>");
        out.println("Subject 3: <input type='number' name='m3'><br><br>");
        out.println("Subject 4: <input type='number' name='m4'><br><br>");
        out.println("Subject 5: <input type='number' name='m5'><br><br>");
        out.println("Subject 6: <input type='number' name='m6'><br><br>");
        out.println("<input type='submit' value='Generate Result'>");
        out.println("</form>");
    }
}
```

OUTPUT:




localhost:8080/Practical_5/Student
localhost:8080/Practical_5/StudentServlet
Welcome, Honey Parmar
Subject 1: 99
Subject 2: 100
Subject 3: 98
Subject 4: 78
Subject 5: 78
Subject 6: 48
Generate Result


localhost:8080/Practical_5/MarksS
localhost:8080/Practical_5/Marks...
Result
Student Name: Honey Parmar
ID: 7894
Department: Phd
FAIL


localhost:8080/Practical_5/MarksS
localhost:8080/Practical_5/MarksServlet
Result
Student Name: Honey Parmar
ID: 7894
Department: Phd
PASS